



AQ8.3: Activity Questions 3 - Not Graded



This assignment will not be graded and is only for practice.

Level 1

1 point

Consider two vectors $a = (1, -2, 2)$, $b = (4, 0, -3)$ $a = (1, -2, 2)$, $b = (4, 0, -3)$. Using the standard dot product as the inner product, the projection of a in the direction of b is given by

☐

$$\left(\frac{-8}{25}, 0, \frac{6}{25}\right)(25-8, 0, 256)$$

☐

$$\left(\frac{-8}{25}, \frac{6}{25}\right)(25-8, 256)$$

☐

$$\left(\frac{8}{25}, 0, \frac{-6}{25}\right)(258, 0, 25-6)$$

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$$\left(\frac{8}{25}, \frac{-6}{25}\right)(258, 25-6)$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\left(\frac{-8}{25}, 0, \frac{6}{25}\right)(25-8, 0, 256)$$

1 point

Consider an orthonormal basis $\left\{\frac{1}{\sqrt{5}}(1, 2), \frac{1}{\sqrt{5}}(-2, 1)\right\}$ of $\mathbb{R}^2$. If $(3, 5) = c_1 v_1 + c_2 v_2$ where $c_1, c_2 \in \mathbb{R}$ and $v_1 = \frac{1}{\sqrt{5}}(1, 2)$, $v_2 = \frac{1}{\sqrt{5}}(-2, 1)$

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$$\frac{1}{\sqrt{5}}(1, 2), \frac{1}{\sqrt{5}}(-2, 1)$$

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$\frac{1}{\sqrt{5}}(-2, 1)$ of $\mathbb{R}^2$. If $(3, 5) = c_1 v_1 + c_2 v_2$ where $c_1, c_2 \in \mathbb{R}$ and $v_1 = \frac{1}{\sqrt{5}}(1, 2)$, $v_2 = \frac{1}{\sqrt{5}}(-2, 1)$

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$$\frac{1}{\sqrt{5}}(1, 2), \frac{1}{\sqrt{5}}(-2, 1)$$

☐

$\frac{1}{\sqrt{5}}(-2, 1)$, then choose the correct option.

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$$c_1 = \frac{13}{\sqrt{5}}, c_2 = \frac{1}{\sqrt{5}} c_1 = \frac{13}{5}$$

☐

$$\frac{13}{\sqrt{5}}, c_2 = \frac{1}{\sqrt{5}}$$

☐

$$c_1 = \frac{-13}{\sqrt{5}}, c_2 = \frac{1}{\sqrt{5}} c_1 = \frac{-13}{5}$$

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$$-\frac{13}{\sqrt{5}}, c_2 = \frac{1}{\sqrt{5}}$$

☐

$$c_1 = \frac{13}{\sqrt{5}}, c_2 = \frac{-1}{\sqrt{5}} c_1 = \frac{-13}{5}$$



$$\sqrt{13}, c_2 = 5$$

$$\sqrt{-1}$$

$$c_1 = \frac{-13}{\sqrt{5}}, c_2 = \frac{-1}{\sqrt{5}}c_1 = 5$$

$$\sqrt{-13}, c_2 = 5$$

$$\sqrt{-1}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$c_1 = \frac{13}{\sqrt{5}}, c_2 = \frac{-1}{\sqrt{5}}c_1 = -5$$

$$\sqrt{13}, c_2 = 5$$

$$\sqrt{-1}$$

1 point

Consider an orthonormal basis $\{(1, 0, 0), (0, 1, 0)\}$ for a subspace W in R^3 . If $x = (1, 2, 3)$ is a vector in R^3 , then which of the following represents a vector in W whose distance from x is the least? Consider dot product as the standard inner product.

$$\circ$$

$$(2, 4, 0)$$

$$\circ$$

$$(3, 4, 0)$$

$$\circ$$

$$(4, 5, 0)$$

$$\circ$$

$$(1, 2, 0)$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$(1, 2, 0)$$

Level 2

Consider the inner product $\langle a, b \rangle = a_1 b_1 - a_1 b_2 - a_2 b_1 + 4a_2 b_2$ where $a = (a_1, a_2)$, $b = (b_1, b_2)$ are vectors in R^2 . Use this information to answer questions 4 and 5.

1 point

Let $x = (1, 2)$. Find the projection of x in the direction of $(3, 4)$ using the inner product defined above.

$$\circ$$

$$\frac{25}{49}(3, 4)$$

$$\circ$$

$$\frac{11}{25}(3, 4)$$

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$\frac{25}{49}(1, 2)$

☐

$\frac{11}{25}(1, 2)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\frac{25}{49}(3, 4)$

1 point

Let $x = (1, 2)$, find the projection of x in a direction perpendicular to $(3, 4)$.

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$\frac{25}{49}(-4, 3)$

☐

$\frac{11}{25}(4, -3)$

☐

$(-\frac{26}{49}, -\frac{2}{49})$

☐

$(\frac{26}{49}, -\frac{2}{49})$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(-\frac{26}{49}, -\frac{2}{49})$

1 point

Consider an orthogonal basis $\{(1, 2, 1), (-2, 0, 2)\}$ of a subspace W , of the inner product space R^3 with respect to the dot product. If $y = (1, 2, 3) \in R^3$, then find $Proj_W(y)$.

☐

$(\frac{1}{3}, \frac{8}{3}, \frac{7}{3})$

☐

$(\frac{18}{\sqrt{6}} - \frac{32}{\sqrt{8}}, \frac{36}{\sqrt{8}}, \frac{18}{\sqrt{6}} + \frac{32}{\sqrt{8}})$

☐

$18 - 8$

☐

$32, 8$

☐

$36, 6$

☐

$18 + 8$

☐

$32)$

☐

$(\frac{1}{3}, -\frac{8}{3}, \frac{7}{3})$

☐

$(\frac{18}{\sqrt{6}} - \frac{32}{\sqrt{8}}, \frac{36}{\sqrt{8}}, -\frac{18}{\sqrt{6}} + \frac{32}{\sqrt{8}})$

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$\sqrt{}$

$$18 - 8$$

 $\sqrt{\quad}$

$$32, 8$$

 $\sqrt{\quad}$

$$36, - 6$$

 $\sqrt{\quad}$

$$18 + 8$$

 $\sqrt{\quad}$

$$32)$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\left(\frac{1}{3}, \frac{8}{3}, \frac{7}{3}\right)(31, 38, 37)$$

[Check Answers](#)

Your score is 0/6